Wi-Fi validation systems

Performance, MLO and coexistence under controlled RF conditions

TECHNICAL CAPABILITIES



MT-SYS **Wi-Fi 7**

Wi-Fi validation coverage

A common test environment connects RF behavior with device performance and application experience.

TEST DOMAIN	EXERCISED CONDITIONS	ENGINEERING OUTPUT
Performance & RF	Throughput, RvR, orientation, client load	Capacity and coverage limits
Wi-Fi 7 & MLO	Link modes, bandwidth, multi-user traffic	Feature behavior and link utilization
Coexistence	Wi-Fi congestion, AWGN, Bluetooth, radar	Robustness under controlled interference
Mobility & mesh	RF transitions, roaming, backhaul load	Continuity and recovery behavior
Applications & QoE	Teams, Zoom, Meet, YouTube, mixed traffic	Latency, loss and application statistics
Functional & debug	Security, association, distributed capture	Protocol evidence for regression and debug

MT-SYS-1 performance test system

A compact rack integrates the RF environment, client radios and test control.

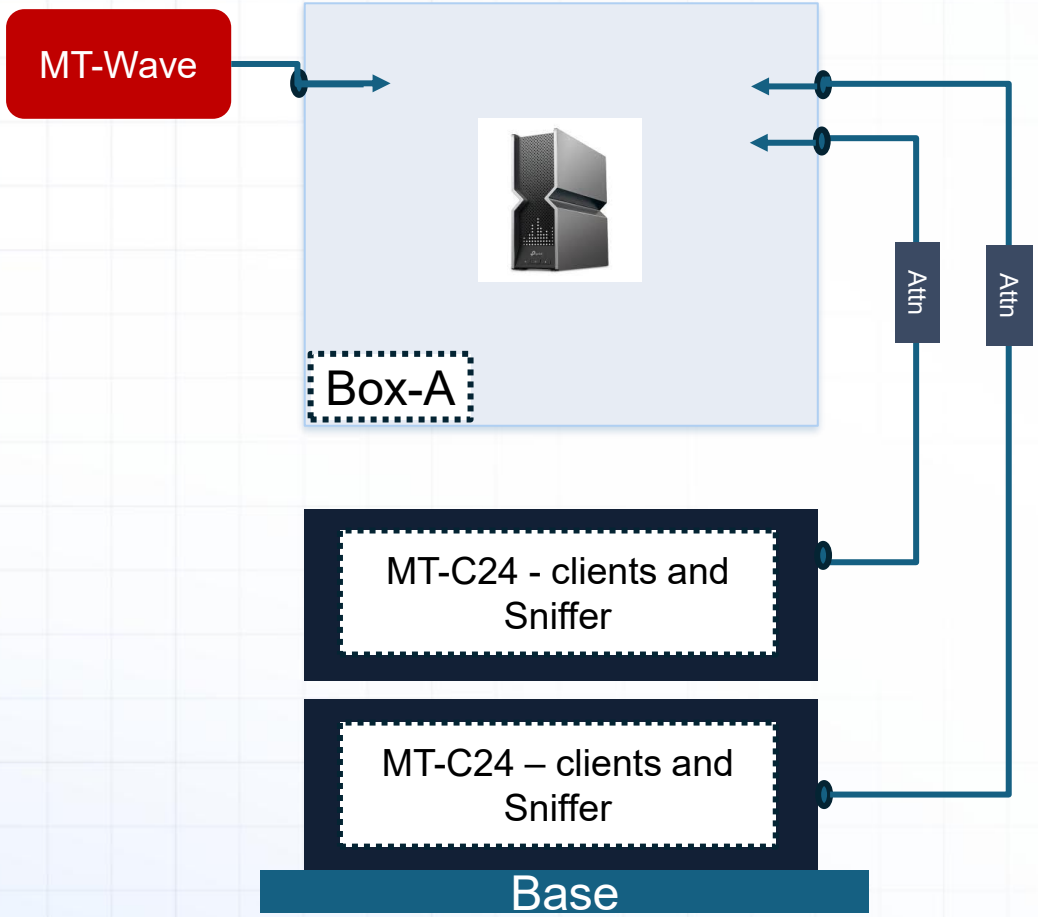
Test Cases

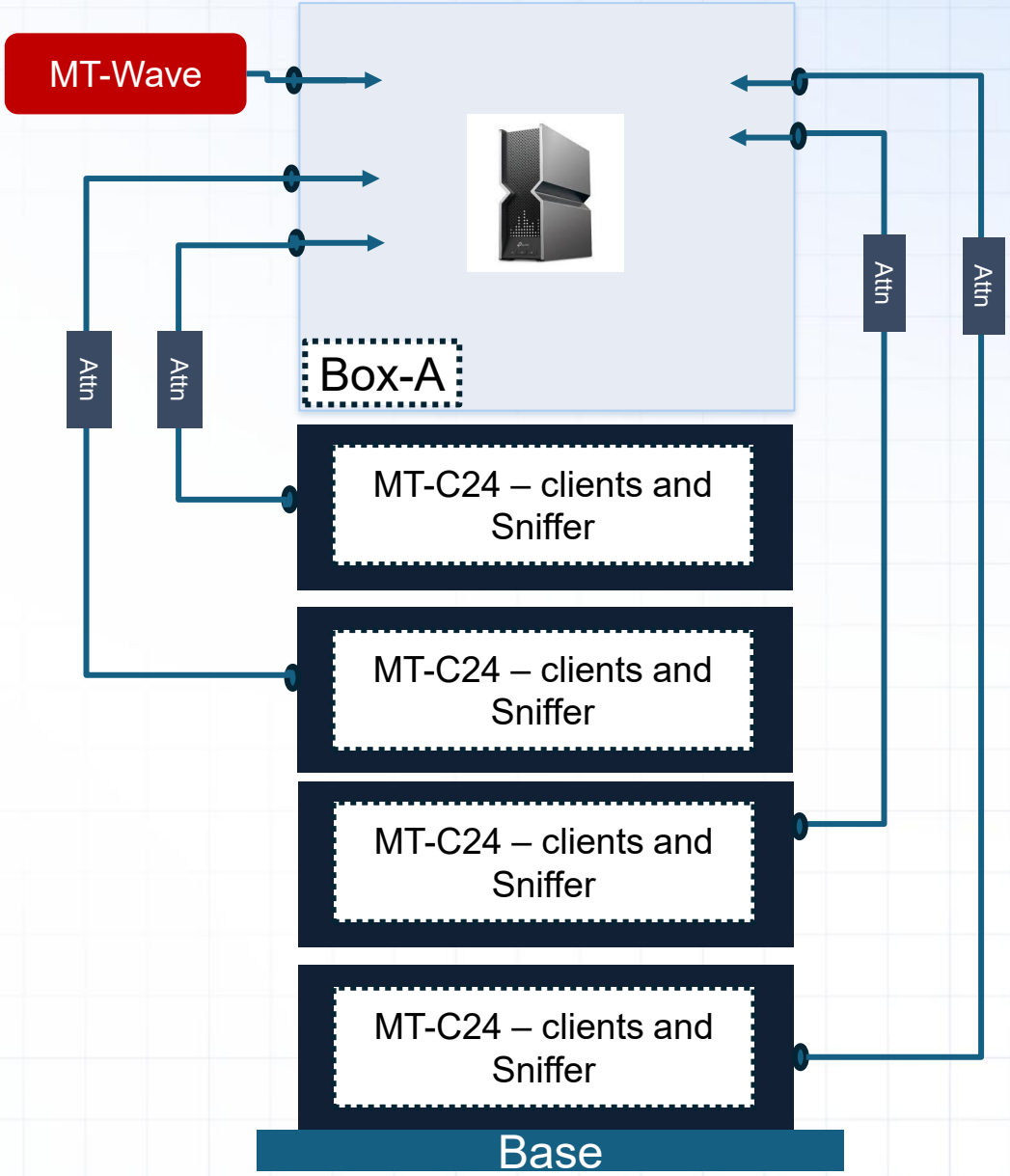
- Peak Performance
- RVR
- RVRVO
- MLO
- Preamble puncturing
- Band steering
- Interference – CCI/ACI
- Congestion control
- Multi-Band throughput
- Multi-client
- Sniffing
- Latency, Jitter
- Encryption and security
- Channel Utilization
- Multi-band sniffing



- 12/24/32 individual radios in single instrument
- Demonstrate highest achievable performance 4K QAM and 320MHz
- MLO – MLMR and eMLSR
- OFDMA, MU-MIMO
- Preamble puncturing
- Multi-client performance test cases
- User Experience – Real streaming – YouTube, Teams, Zoom, Google Meet and many more

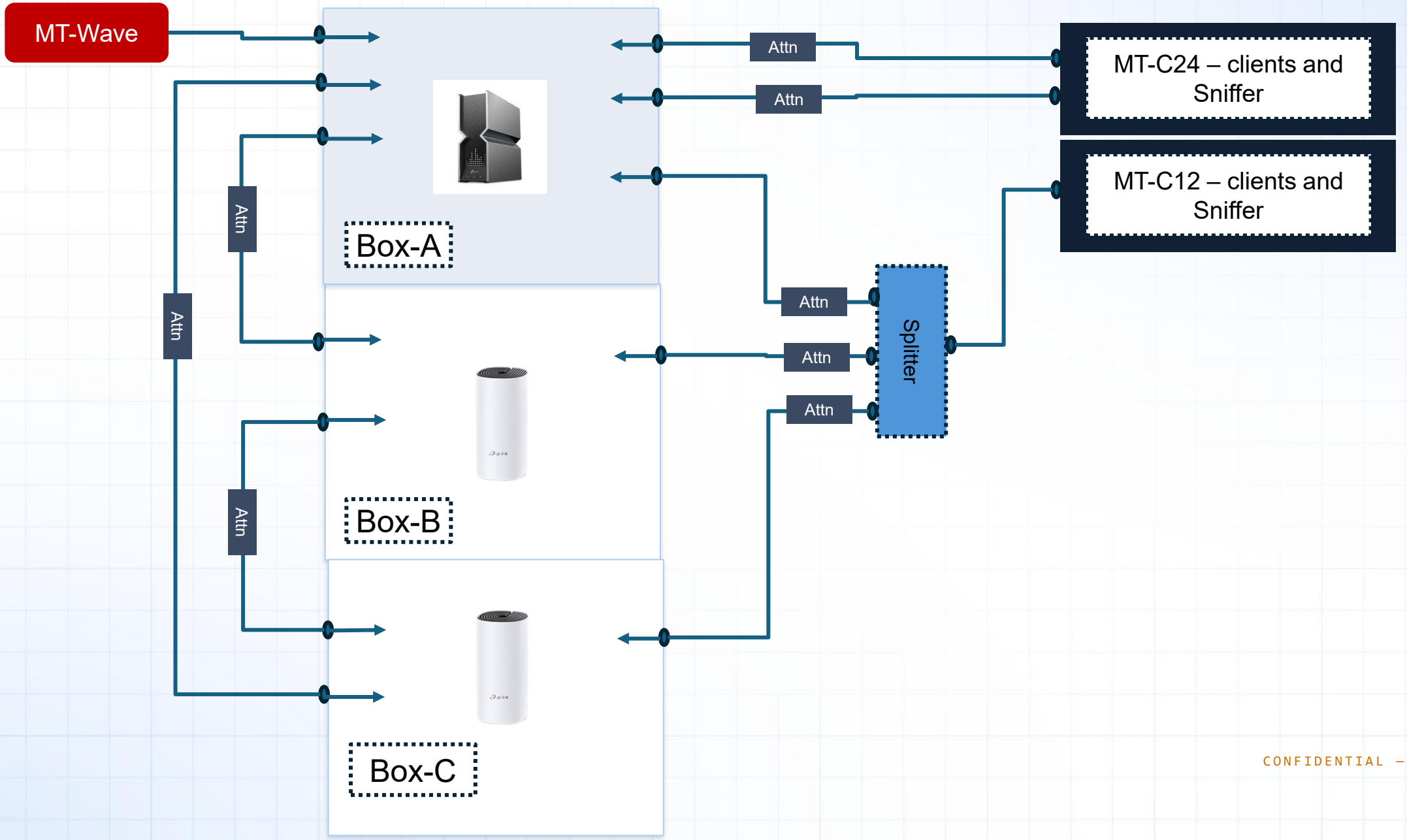
24 / 48 / 96 Stations — Scalable





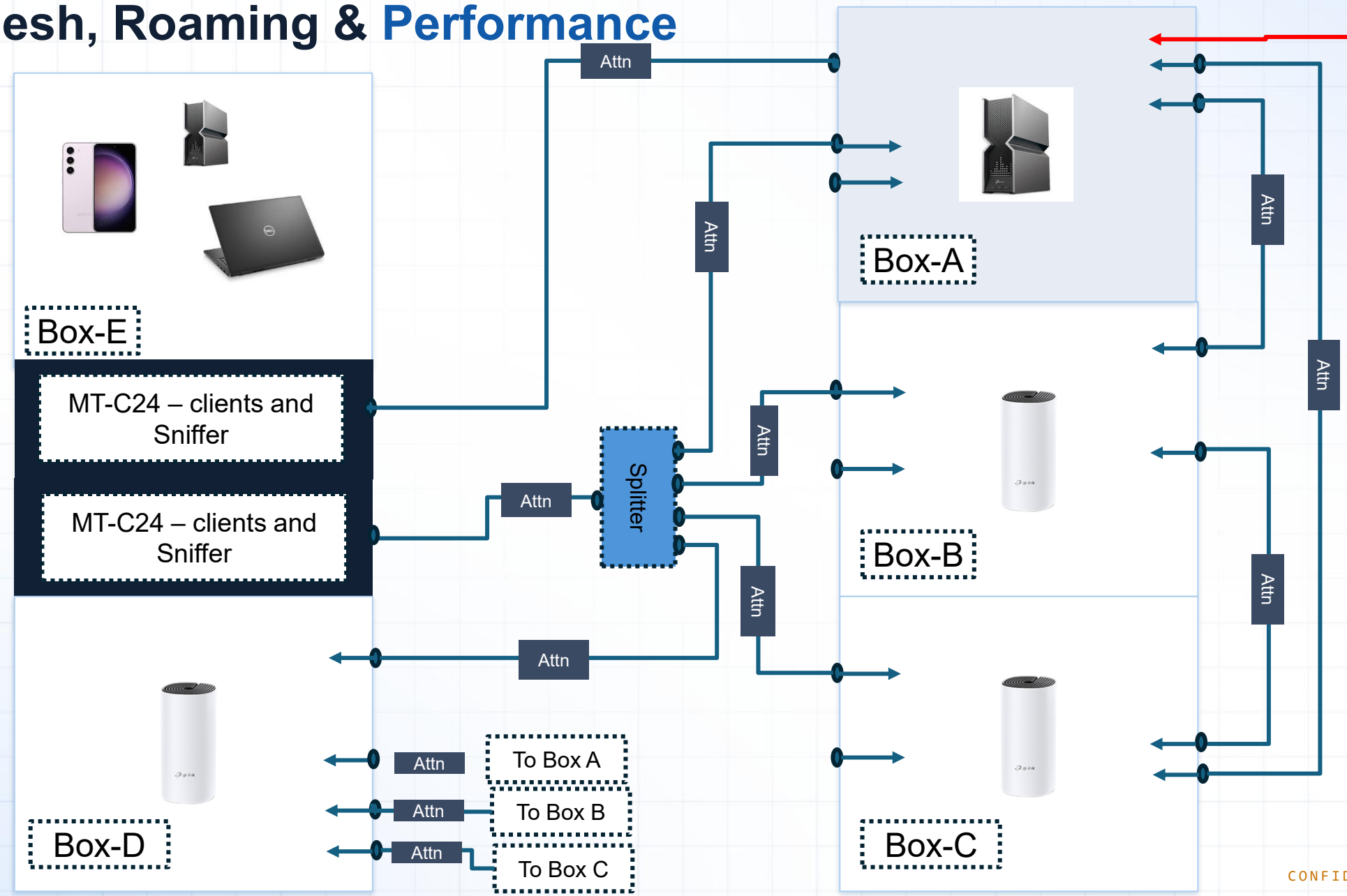
CONFIDENTIAL – SHARED UNDER NDA

Performance & TR-398 Test System



CONFIDENTIAL – SHARED UNDER NDA

Mesh, Roaming & Performance



CONFIDENTIAL – SHARED UNDER NDA

A modular platform for different lab roles

PLATFORM	ROLE IN THE TEST SYSTEM	TYPICAL EVALUATION
MT-SYS	Rack and shielded RF environment	AP/CPE qualification and regression
MT-C24 / MT-C4	Real-radio clients with individual RF paths	Client scaling, MLO and interoperability
MT-BE	Up to 4×4 MIMO emulation	High-throughput and multi-link testing
MT-Wave	Wi-Fi and non-Wi-Fi stimulus	Congestion, coexistence and DFS
RF attenuators	Controlled loss on each RF path	Rate vs range, steering and roaming
MT-Core + MT-Sniffer	Automation, HTML reports and capture	Regression with debug evidence

For network operators, device brands, chipset teams, ODMs and independent test laboratories.

Client emulator configurations

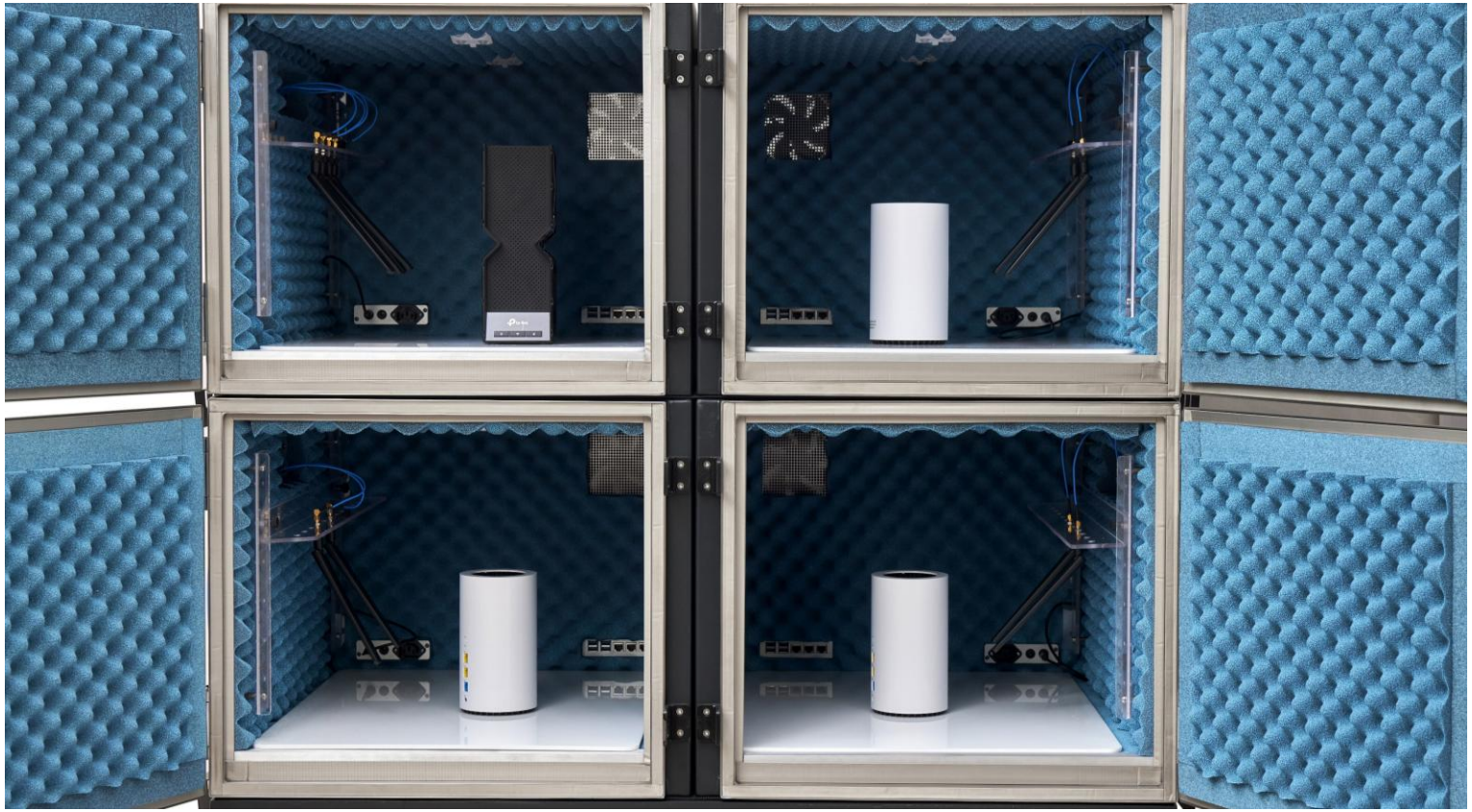
Platform selection follows spatial streams, station count and the MLO behavior under test.

CAPABILITY	MT-C24 / MT-C4	MT-BE
Radio architecture	Up to 2×2 MIMO per radio MT-C24: up to 24 radios	Up to 4×4 MIMO
Station scale	Scales through real-radio modules	Up to 128 stations per band
MLO modes listed	eMLSR, MLMR, MLSR	MLMR, STR
Channel widths listed	80 / 160 / 320 MHz	80 / 160 / 320 MHz
AP and STA roles	Supported configurations	Supported configurations
Multi-user features	OFDMA, MU-MIMO, puncturing	OFDMA, MU-MIMO, puncturing
RF and visibility	Individual shielded paths and sniffing	Individual shielded paths and monitoring

Feature availability depends on the selected radio, OS/driver, DUT and region. Peak results appear separately.

RF isolation supports repeatable OTA tests

Shielded chambers contain the DUT and help control external signals and internal reflections.



>90 dB isolation

Specified RF isolation up to 8 GHz.

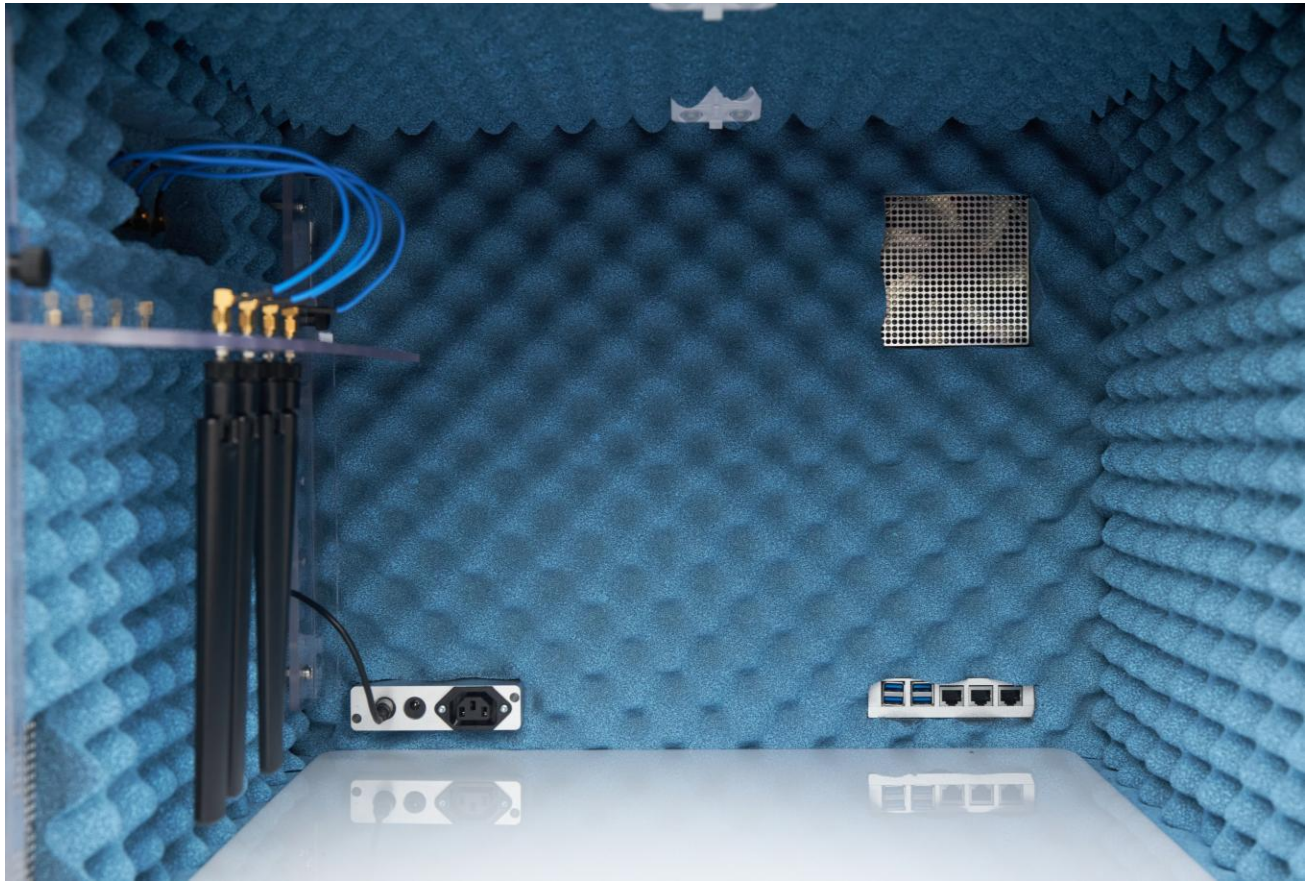
Absorber-lined interiors

Convoluted and flat absorbers support controlled OTA conditions.

Integrated I/O

10G Ethernet, USB and up to 24 SMA connections in the listed configuration.

Individual RF paths make tests controllable



A path for each radio

Dedicated RF chains allow independent attenuation and observation of each test link.

Antenna selection

Omnidirectional and directional antennas support different traffic and coverage scenarios.

Path-loss characterization

Cable loss, splitters and attenuators form part of the test reference plane.

MT-Core test automation

Test setup, execution and reporting use a common sequence and configurable limits.

01

Define

Select test cases
Set traffic and RF profiles
Choose acceptance limits

02

Execute

Apply DUT and client settings
Run the test sequence
Monitor per-test status

03

Evaluate

Compare KPIs with limits
Capture diagnostic context
Record the verdict

04

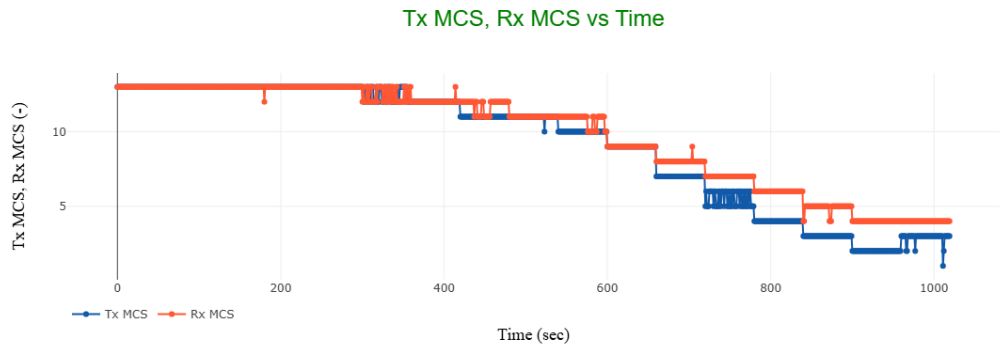
Report

Export an HTML report
Review KPIs and charts
Share the test evidence

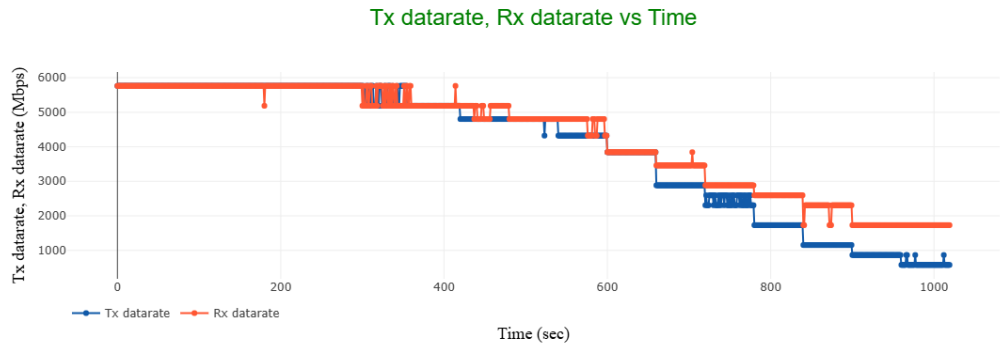
Automation support and DUT command integration follow the configured test package.

RF behavior in live telemetry

TX / RX MCS over time



TX / RX PHY data rate over time



Per-link visibility

Data and beacon RSSI, MCS, channel width, rate, packets, retries and failures.

MLO session context

Bands, channels and per-band traffic identify how the connection uses its links.

Metric availability depends on the client and capture metadata. One-way latency requires an appropriate timing method.

Measured multi-gigabit throughput

>5.4

Gbps, MT-C24
2×2 MIMO OTA

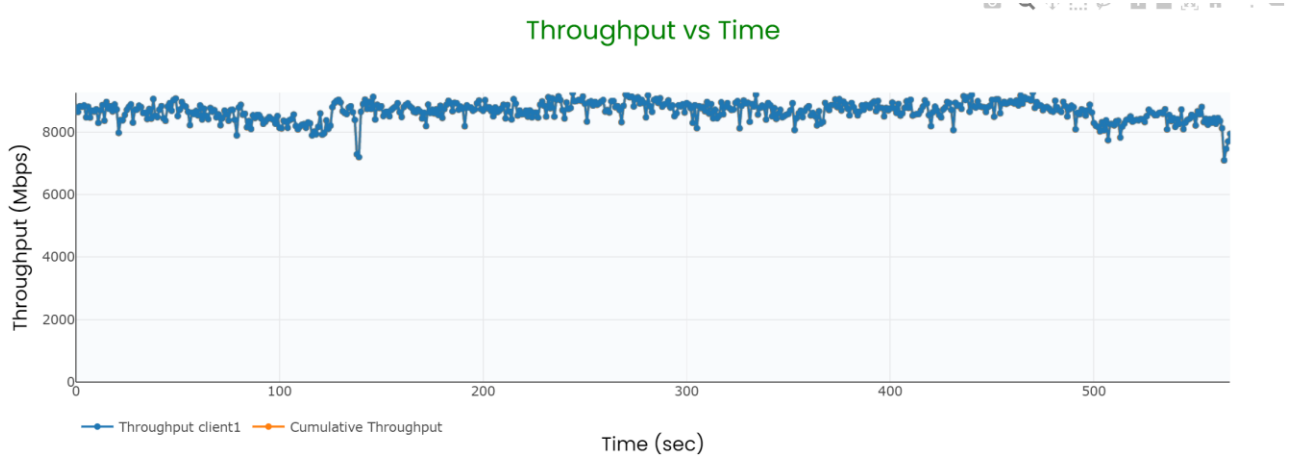
>8.5

Gbps, MT-BE
TCP MLO OTA

>9.5

Gbps
Reported UDP peak

MT-BE TCP MLO throughput trace

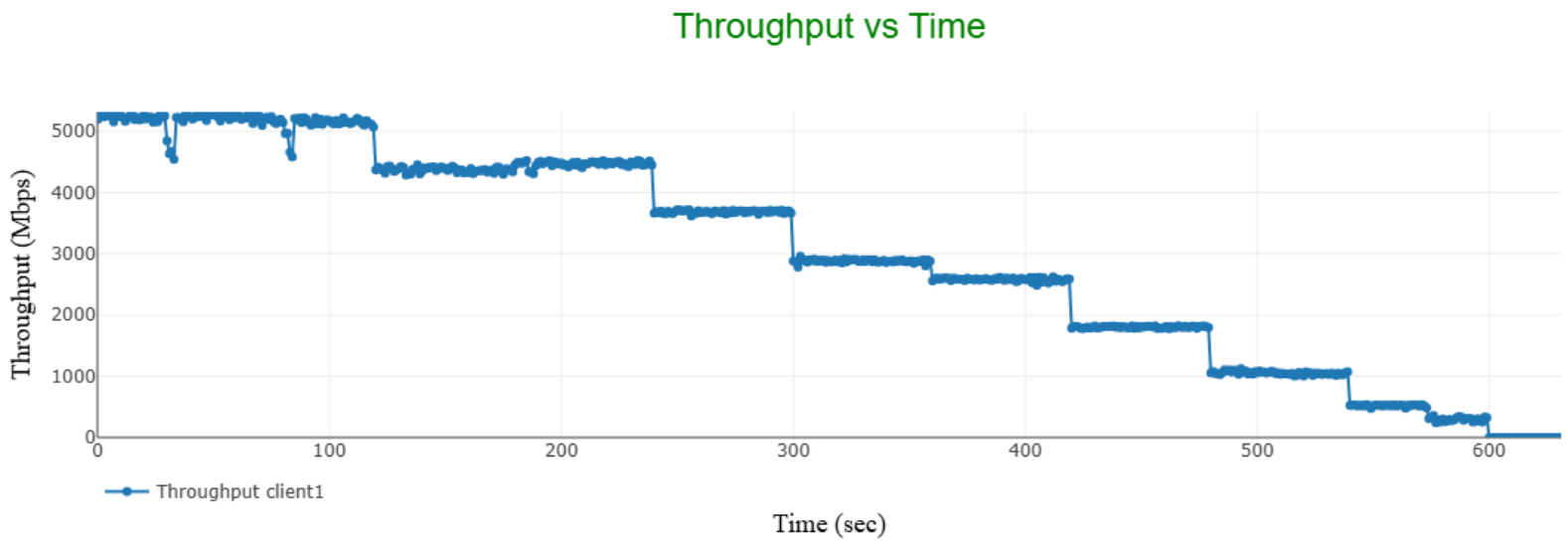


Wavemetrik lab examples. Throughput depends on the DUT, bands, bandwidth, RF path and traffic settings.

Rate adaptation as attenuation increases

The throughput curve identifies operating regions and the onset of link degradation.

MT-C24 throughput over time



Controlled variable

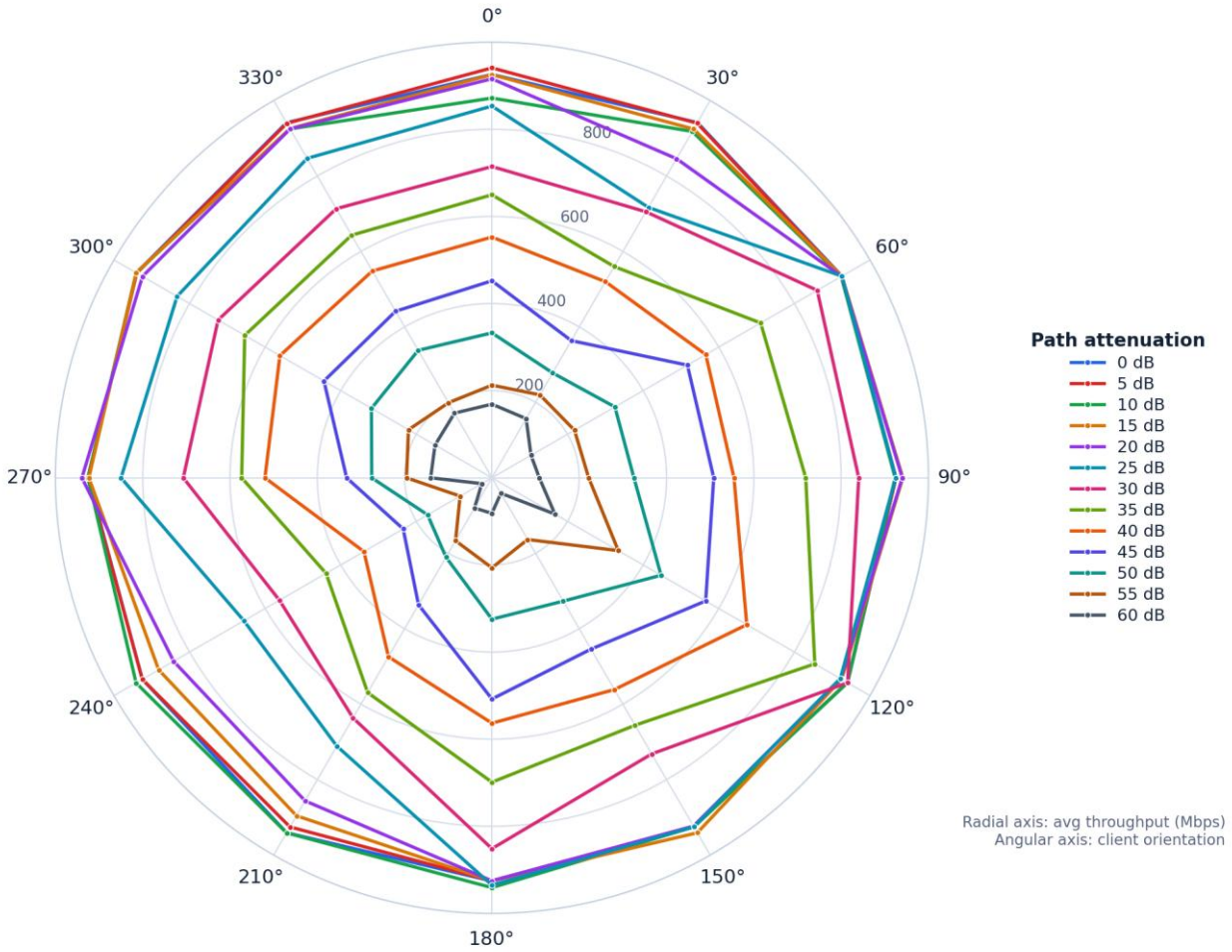
Increase attenuation in steps while maintaining the selected traffic profile.

Correlated evidence

Review throughput alongside RSSI, MCS, retries and receive drops.

Horizontal axis: elapsed time. The trace shows a sweep, rather than sustained peak throughput over its entire duration.

Rate vs range across device orientations



942

Mbps peak at 0 dB attenuation

Orientation sensitivity

The reported spread at 0 dB is approximately 2%.

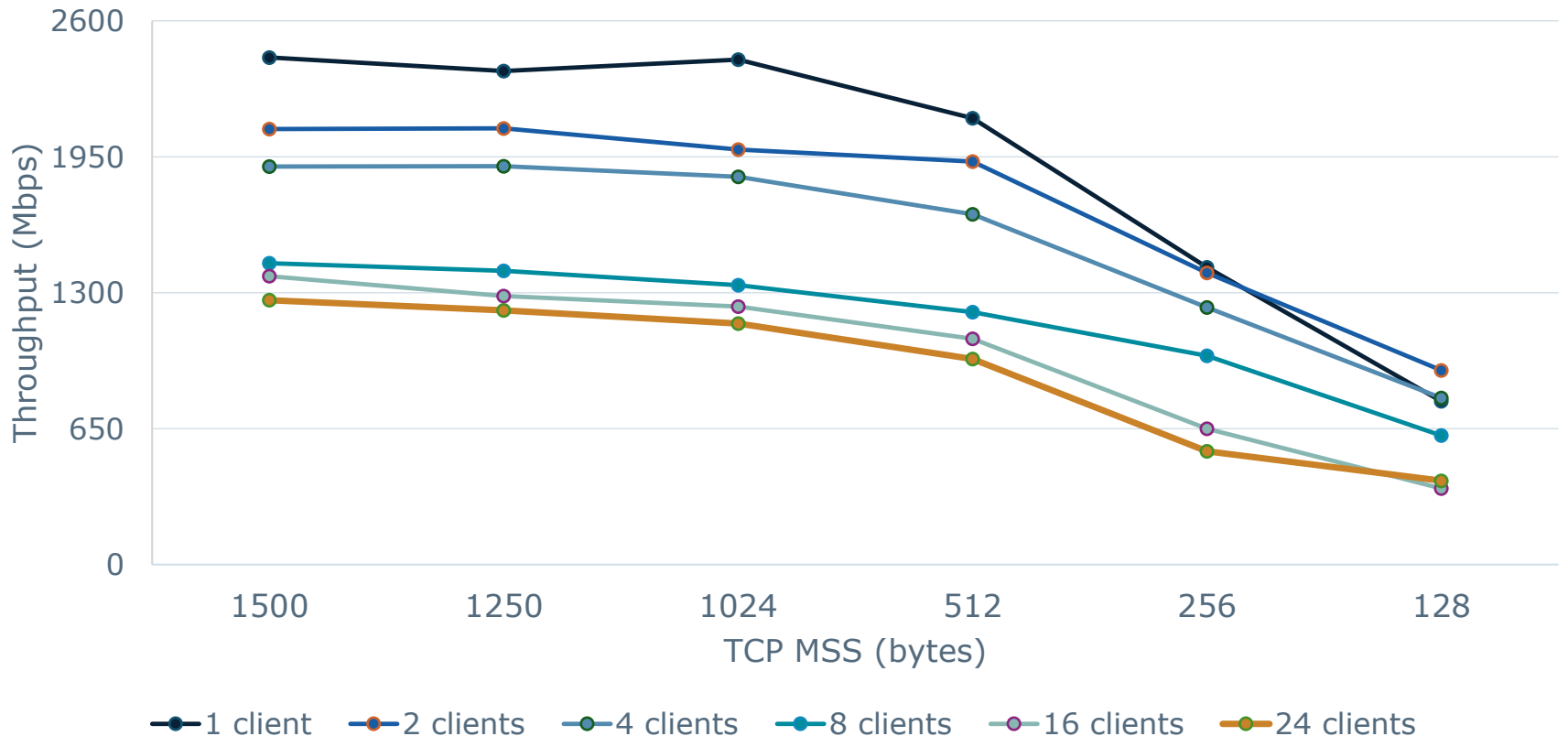
Coverage roll-off

Mean throughput: 907 Mbps at 20 dB, 536 Mbps at 40 dB and 116 Mbps at 60 dB.

Radius: average throughput. Angle: device orientation. Each trace represents one attenuation setting.

Client load and packet size affect capacity

Aggregate TCP throughput for 1, 2, 4, 8, 16 and 24 clients in an example MLO test.



2,425

Mbps with 1 client at 1500-byte MSS

1,264

Mbps aggregate with 24 clients at the same MSS

Test configuration: 2.4 GHz at 40 MHz and 6 GHz at 160 MHz. Measurements show aggregate TCP throughput.

Wi-Fi 7 PHY and channel-width coverage

BAND	CHANNEL WIDTHS	VALIDATION FOCUS
2.4 GHz	20 / 40 MHz	Interoperability and coexistence
5 GHz	20 / 40 / 80 / 160 MHz	Capacity, MLO links and DFS behavior
6 GHz	20 / 40 / 80 / 160 / 320 MHz	Wide-channel performance and MLO

4096-QAM and MCS

Characterize throughput and rate behavior at controlled RF levels.

Multi-user operation

Exercise OFDMA, MU-MIMO, beamforming and Multi-RU combinations.

320 MHz applies to 6 GHz. Available channels and feature combinations depend on the region, DUT and client implementation.

MLO operating modes and test objectives

Mode negotiation and packet timing determine which multi-link behavior a run demonstrates.

MODE / PROPERTY	BEHAVIOR TO EXAMINE	TEST EVIDENCE
eMLSR	Link selection and switching with one active data exchange at a time	Negotiated mode, link changes and interruption time
MLMR	Multiple radios operating across affiliated links	Per-link traffic, aggregate throughput and cross-link activity
STR	Simultaneous transmit and receive across an eligible link pair	Negotiated capability and overlapping TX/RX activity for the same client

MLMR describes a radio arrangement. STR describes a simultaneous operating capability. Device support and link-pair constraints apply.

Multi-band capture in one PCAPNG

Dedicated sniffer radios observe the affiliated links while retaining channel and interface identity.

o.	Time	Source	Dest	Protocol	Length	Info	New Column	data rate	TX rate	RSSI	DSCP	packet type
269274	49.435772099	sunwe…	Hu…	802.11	1630	QoS Data, SN=3390, FN=0, Flags=.p....F.C	5200 MHz	1729.400…		-44 dBm		2
269275	49.435774871	sunwe…	Hu…	802.11	1630	QoS Data, SN=3391, FN=0, Flags=.p....F.C	5200 MHz	1729.400…		-44 dBm		2
269276	49.435776805	sunwe…	Hu…	802.11	1630	QoS Data, SN=3392, FN=0, Flags=.p....F.C	5200 MHz	1729.400…		-44 dBm		2
269277	49.435778174	sunwe…	Hu…	802.11	1630	QoS Data, SN=3393, FN=0, Flags=.p....F.C	5200 MHz	1729.400…		-44 dBm		2
269278	49.435779906	sunwe…	Hu…	802.11	1630	QoS Data, SN=3394, FN=0, Flags=.p....F.C	5200 MHz	1729.400…		-44 dBm		2
269279	49.435781569	sunwe…	Hu…	802.11	1630	QoS Data, SN=3395, FN=0, Flags=.p....F.C	5200 MHz	1729.400…		-44 dBm		2
269280	49.435783137	sunwe…	Hu…	802.11	1630	QoS Data, SN=3396, FN=0, Flags=.p....F.C	5200 MHz	1729.400…		-44 dBm		2
269281	49.435784337	sunwe…	Hu…	802.11	1630	QoS Data, SN=3397, FN=0, Flags=.p....F.C	5200 MHz	1729.400…		-44 dBm		2
269282	49.435833771	HuiZh…	su…	802.11	326	QoS Data, SN=1082, FN=0, Flags=.p.....TC	2412 MHz	229.3999…		-12 dBm		2
269283	49.435836406	HuiZh…		802.11	568	QoS Data, SN=1083, FN=0, Flags=.p.....TC	2412 MHz	229.3999…		-12 dBm		2
269284	49.435838797	HuiZh…		802.11	568	QoS Data, SN=1084, FN=0, Flags=.p.....TC	2412 MHz	229.3999…		-12 dBm		2
269285	49.435840453	HuiZh…		802.11	520	QoS Data, SN=1085, FN=0, Flags=.p.....TC	2412 MHz	229.3999…		-12 dBm		2
269286	49.435841929	HuiZh…		802.11	532	QoS Data, SN=1086, FN=0, Flags=.p.....TC	2412 MHz	229.3999…		-12 dBm		2
269287	49.435843592	HuiZh…		802.11	532	QoS Data, SN=1087, FN=0, Flags=.p.....TC	2412 MHz	229.3999…		-12 dBm		2
269288	49.435845082	HuiZh…		802.11	532	QoS Data, SN=1088, FN=0, Flags=.p.....TC	2412 MHz	229.3999…		-12 dBm		2
269289	49.435846987	HuiZh…		802.11	532	QoS Data, SN=1089, FN=0, Flags=.p.....TC	2412 MHz	229.3999…		-12 dBm		2
269290	49.435848366	HuiZh…		802.11	520	QoS Data, SN=1090, FN=0, Flags=.p.....TC	2412 MHz	229.3999…		-12 dBm		2

Chronological merging uses captured timestamps. Cross-radio timing accuracy depends on the capture clocks and synchronization method.

Link identity and timing in MLO debug

DEBUG QUESTION	EVIDENCE TO CORRELATE	PRACTICAL OUTCOME
Do the links belong to one MLD?	Multi-Link elements and affiliated AP/client addresses	Separate one MLO session from unrelated clients on the same SSID
How does traffic move between links?	Per-link data, management events and channel/RSSI context	Locate a link switch, load change or recovery event
Does the client exchange concurrently?	Overlapping same-client activity and validated capture timing	Distinguish activity on two bands from demonstrated simultaneous operation
What changed at the failure?	Test ID, DUT state, RF profile and packet timestamp	Share a reproducible vendor debug case

Wireshark filters use the link BSSID and captured channel. One SSID alone does not identify an MLO session.

Puncturing and multi-user validation

Feature testing combines a controlled stimulus with evidence from actual transmissions.

FEATURE	CONTROLLED STIMULUS	PACKET / PHY EVIDENCE	SERVICE KPI
Preamble puncturing	Secondary-subchannel interference	EHT Operation signaling and EHT PPDU metadata	Throughput and active bandwidth
OFDMA	Multiple active clients and selected traffic mix	Trigger/RU allocation and multi-user PPDUs	Latency, capacity and per-client fairness
MU-MIMO	Concurrent spatially separated clients	Sounding and multi-user spatial transmissions	Aggregate and per-client throughput

Advertised capability and observed feature use answer different questions.

MT-Wave interference generator

Programmable RF stimulus for coexistence, congestion and robustness testing.



+20 to −30

dBm programmable output range

Level control

1 dB steps with specified ±0.5 dB accuracy.

Test integration

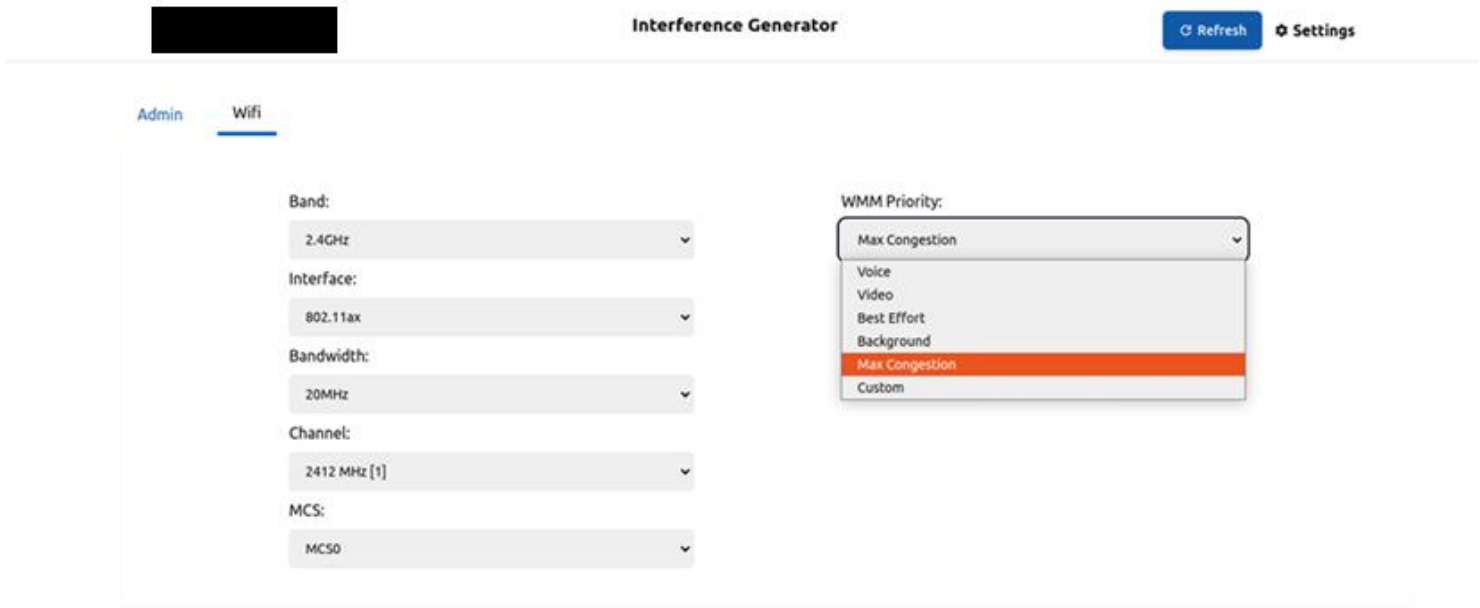
Browser control and Ethernet / REST interfaces support manual and automated operation.

2.4 / 5 / 6 GHz coverage applies to supported signal profiles and the selected instrument configuration.

Controlled Wi-Fi traffic and congestion

Packet parameters, spacing and priority define the applied traffic condition.

Channel, bandwidth and traffic configuration



Frame-level controls

Select PHY settings, payload, output level and WMM access category.

Timing controls

Configure inter-frame spacing within the specified 16–400 µs range.

Measured outcome

Record the actual busy-time metric and DUT KPIs alongside each generator setting.

Interference profiles and failure modes

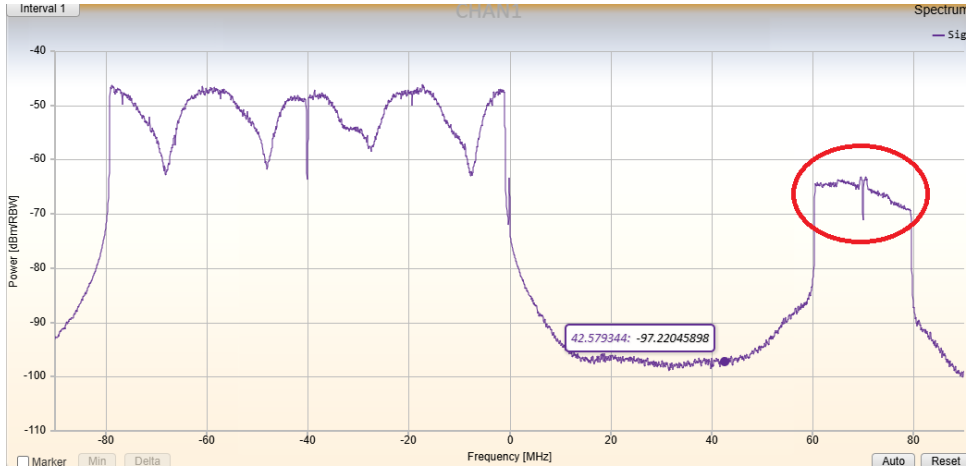
SIGNAL FAMILY	SIGNAL PROFILES	USE IN THE LAB
Wi-Fi / OFDM	Packet traffic and 80 MHz OFDM interferers	Co-channel and adjacent-channel load
Noise	Wideband AWGN, including a 20 MHz example	Receiver desense and blocking
Bluetooth	Classic BR, EDR 2/3 Mbps, BLE 1M/2M/Coded	2.4 GHz coexistence
IEEE 802.15.4	Zigbee O-QPSK with half-sine shaping	IoT coexistence
DFS radar	FCC types 1–6 and ETSI types 1–6	Radar detection and channel response

Select profiles by operating band and installed waveform package. Representative instrument captures appear in the appendix.

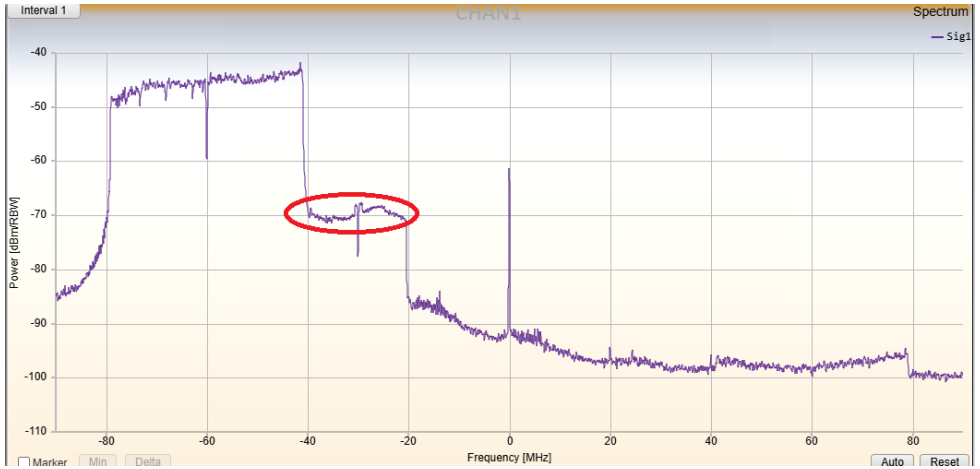
Co-channel and adjacent-channel interference

Interferer placement and power become explicit variables in the evaluation.

Interferer near the channel edge



Interferer within the channel span



Configuration record

Wanted channel, interferer frequency, bandwidth and level at the DUT reference plane.

DUT response

Throughput, packet loss, latency, retries and any bandwidth or channel change.

DFS radar testing and channel response

PROFILE SET	SIGNALS LISTED	RESPONSE TO MEASURE
FCC	Types 1–4: short pulse Type 5: long-pulse chirp Type 6: frequency hopping	Detection and channel response under the applicable test procedure
ETSI	Types 1–3: constant PRF Type 4: chirp Types 5–6: staggered PRF	Detection and channel response under the applicable test procedure

Stimulus validation

Characterize frequency, pulse width, timing, burst structure and power at the DUT.

Regulatory pre-test

Use the selected region and standard edition to define timing and pass criteria.

Real application traffic under controlled RF

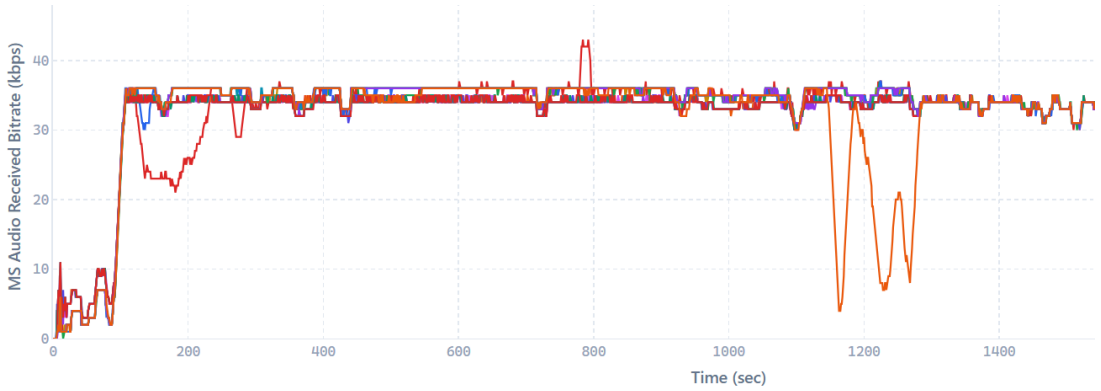
Application sessions add user-experience context to throughput and packet measurements.

WORKLOAD	APPLICATION STATISTICS	RF / NETWORK CONDITIONS
Microsoft Teams	Audio/video bitrate, jitter, RTT, loss and MOS	Attenuation, congestion and client load
Zoom	Audio bitrate, jitter, RTT and MOS	Stable RF and changing link conditions
YouTube	Playback frame rate, buffer and bitrate	Long-duration streaming and coexistence
Google Meet	Application/session statistics as available	Live traffic in the selected test setup
Mixed applications	Per-application behavior and traffic distribution	Competing WMM priorities and RF transitions

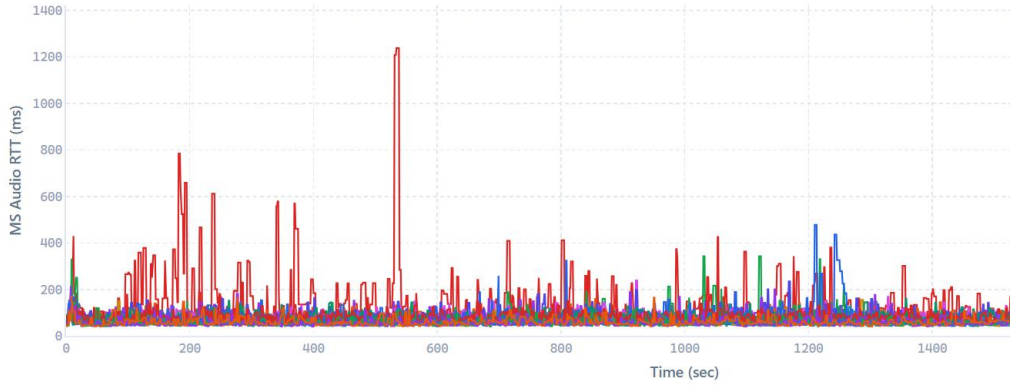
Available KPIs depend on the application, client and telemetry source. Example Teams, YouTube and Zoom plots follow.

Microsoft Teams media quality

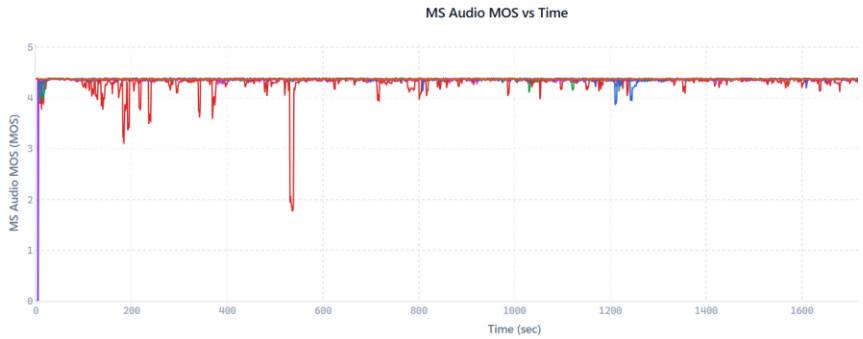
Audio bitrate



Audio round-trip time



Audio MOS

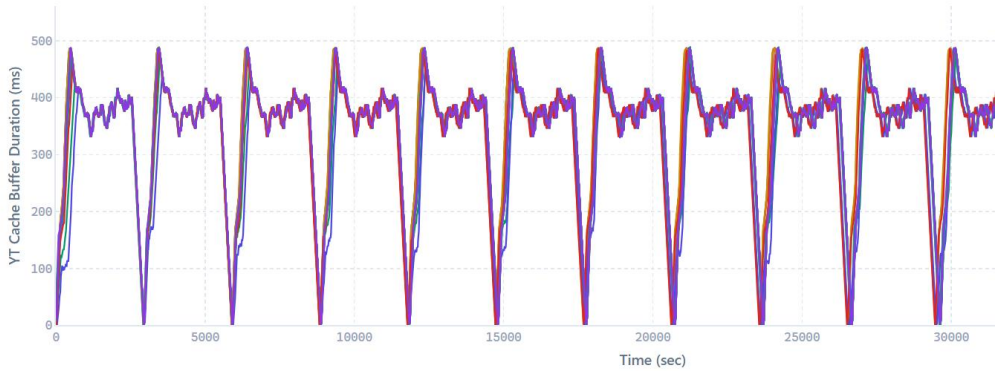


Engineering use

Correlate media adaptation and quality excursions with the applied RF profile and network events.

YouTube buffer and playback continuity

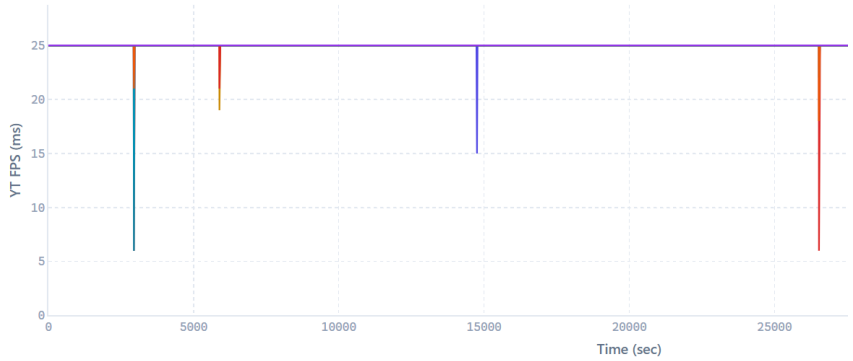
Playback buffer over time



Playback continuity

Identify buffer depletion, playback interruptions and frame-rate changes.

Playback frame rate over time

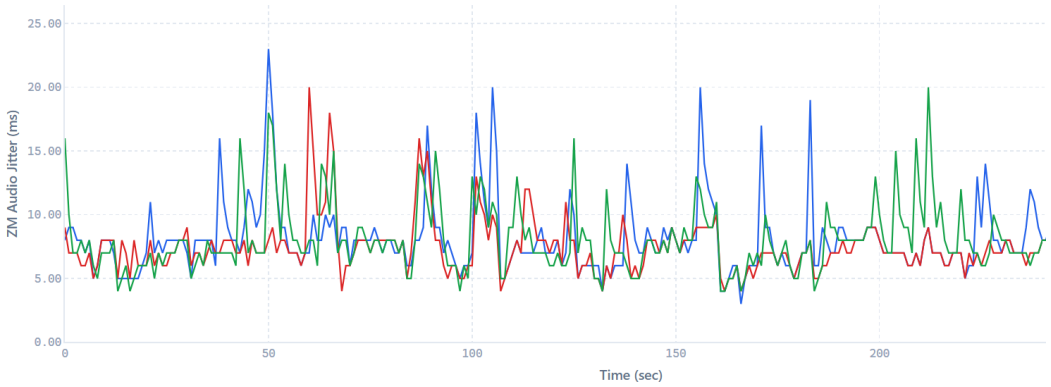


Extended runs

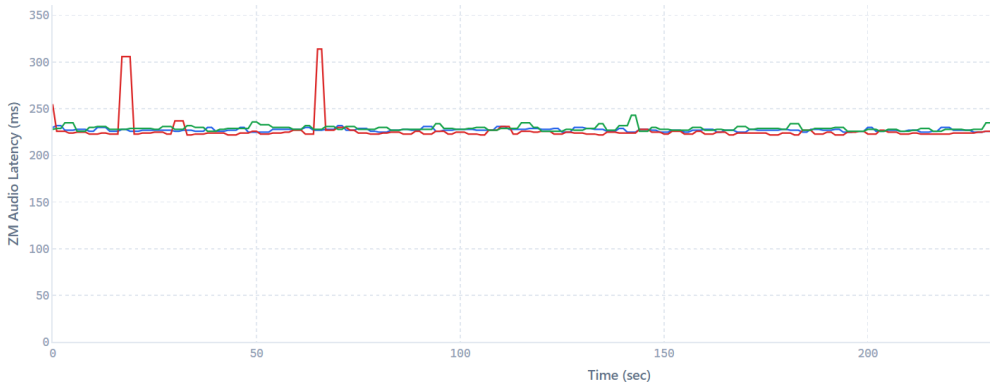
Evaluate recurring patterns across the complete session and the recorded RF timeline.

Zoom: audio stability and latency

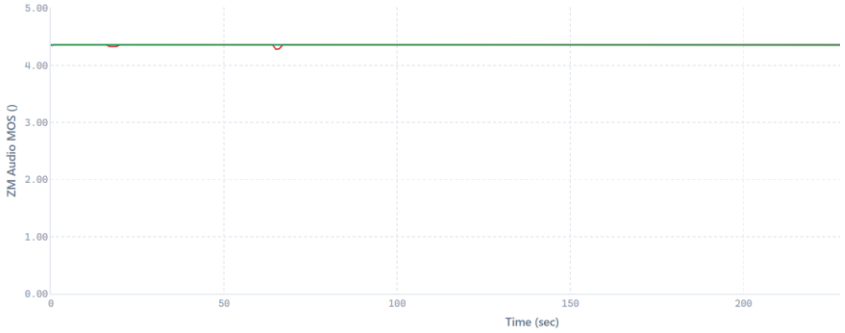
Audio jitter



Audio round-trip time



Audio MOS

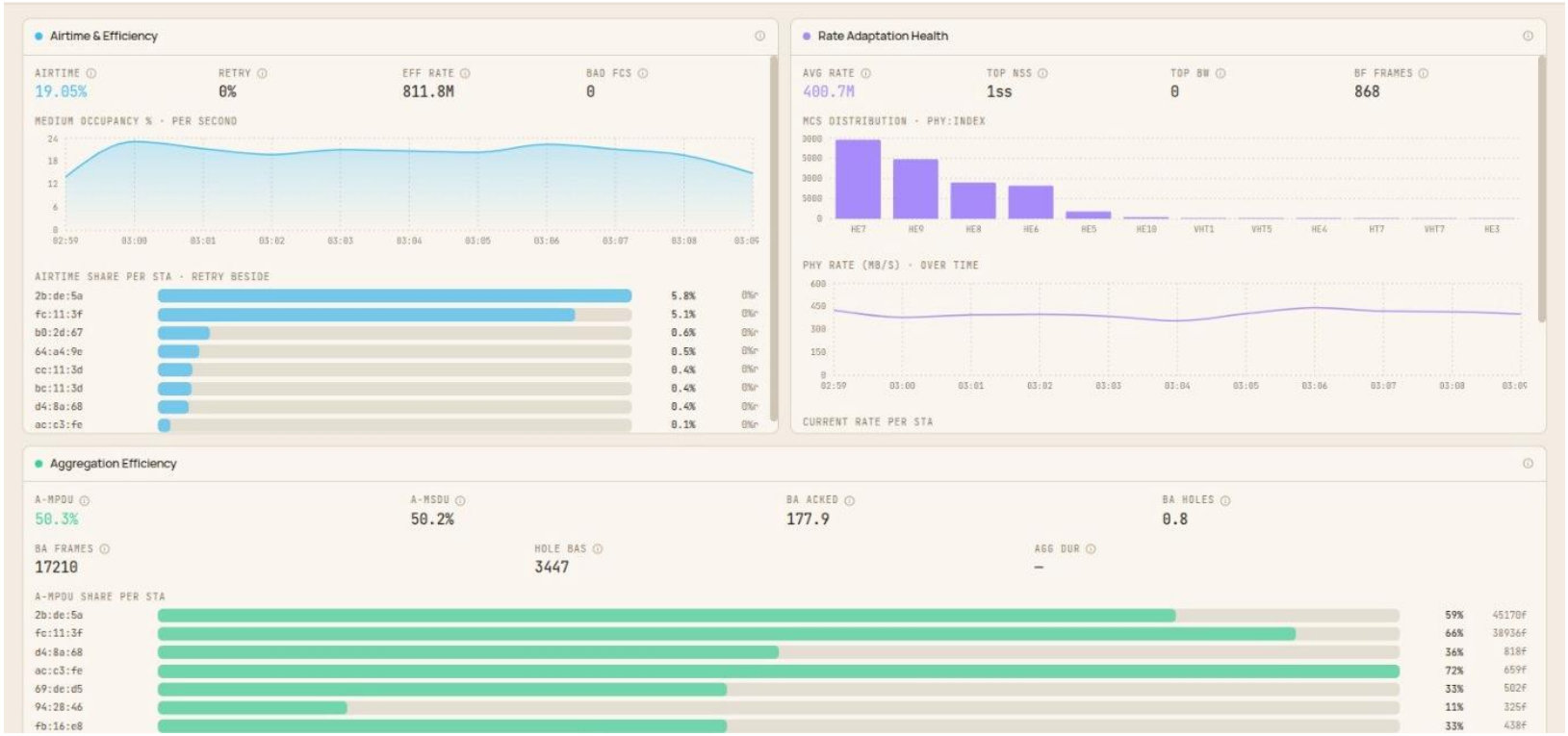


Engineering use

Track jitter and latency excursions while checking whether the application sustains audio quality.

MT-Sniffer distributed capture and analysis

A browser console coordinates capture benches and presents live station-level diagnostics.



Central orchestration

Coordinate capture starts, stops and test identifiers across benches.

Per-STA analysis

Review airtime estimates, RSSI, rates and retries with frame context.

Captured data and decoded features depend on the radio, driver and supplied PHY metadata.

Nine modules for packet-level diagnosis

PERFORMANCE & ACCESS	PHY & MULTI-USER	PROTOCOL & STABILITY
Airtime & efficiency Occupancy estimates and per-STA share	Rate adaptation MCS, NSS and PHY rate	QoS / WMM / EDCA Access categories and TIDs
Aggregation A-MPDU, A-MSDU and Block Ack	OFDMA / MU-MIMO Triggers, RUs and multi-user frames	Connection stability Association, deauthentication and reasons
Channel & contention RTS/CTS and NAV reservations	Signal quality & PHY RSSI and available noise / SNR metadata	Frame inspector Management, control and data frames

Airtime estimates, NAV reservations and physical CCA busy time describe different measurements. Noise and SNR require valid capture metadata.

TR-398 and extended test coverage

The selected standard edition and customer requirements define the executable case matrix.

COVERAGE GROUP	TEST CASES
Baseline performance	Maximum TCP throughput, receiver sensitivity, rate vs range, spatial consistency
Multi-station and traffic	Maximum connections, fairness, multi-station, bidirectional, multi-band and multicast
Stability and coexistence	Association stability, long-term stability, AP coexistence and channel selection
Quality and multi-user	QoS, latency under load, DL/UL MU-MIMO and OFDMA
Wi-Fi 7 and mobility	MLO eMLSR / STR, puncturing, mesh RvR and roaming

Coverage and limits follow the agreed case matrix, equipment configuration and standard edition. Formal certification is a separate process.

A focused DUT evaluation

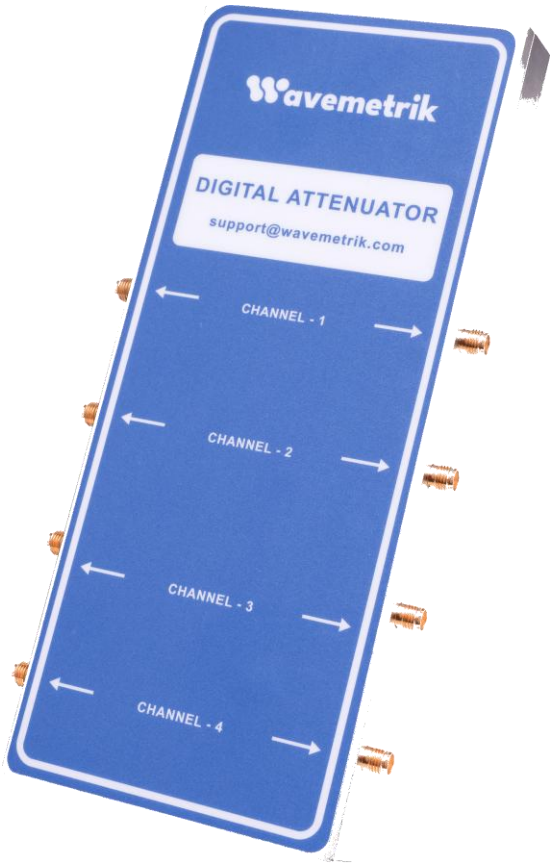
A proposed sequence establishes the baseline before adding feature and robustness tests.

STAGE	TEST ACTIVITY	REVIEWABLE OUTPUT
01 Bring-up	Confirm DUT firmware, radio settings and RF path calibration	Recorded configuration and baseline trace
02 Performance	Run single-link and MLO throughput with an attenuation sweep	Throughput/RSSI/MCS behavior and selected acceptance limits
03 Robustness	Add client load, interference and the required application traffic	Latency, loss, recovery and application statistics
04 Debug and review	Collect per-link capture evidence and rerun selected cases	HTML results and a reproducible issue record for engineering review

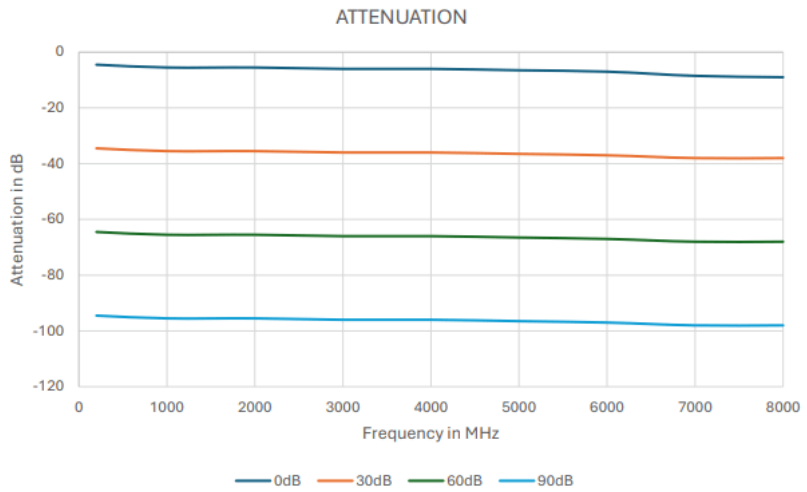
The final configuration and acceptance criteria are agreed against the DUT and intended use cases.

Programmable attenuator characteristics

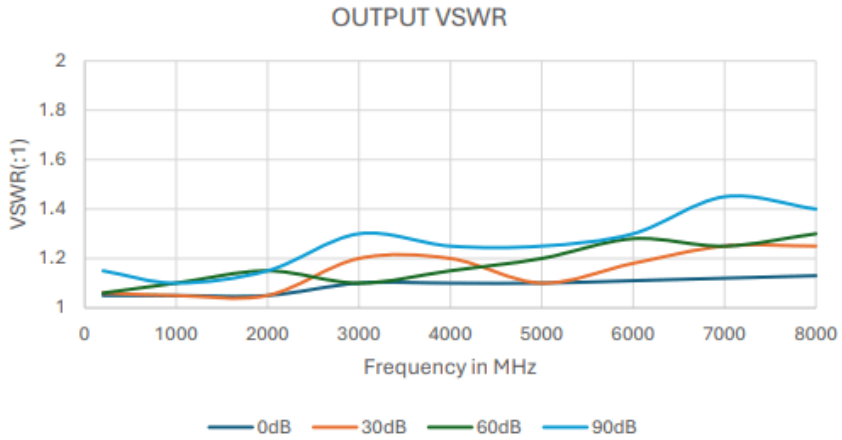
200 MHz–8 GHz operation with up to 95 dB attenuation and 0.25 dB steps.



Attenuation vs frequency



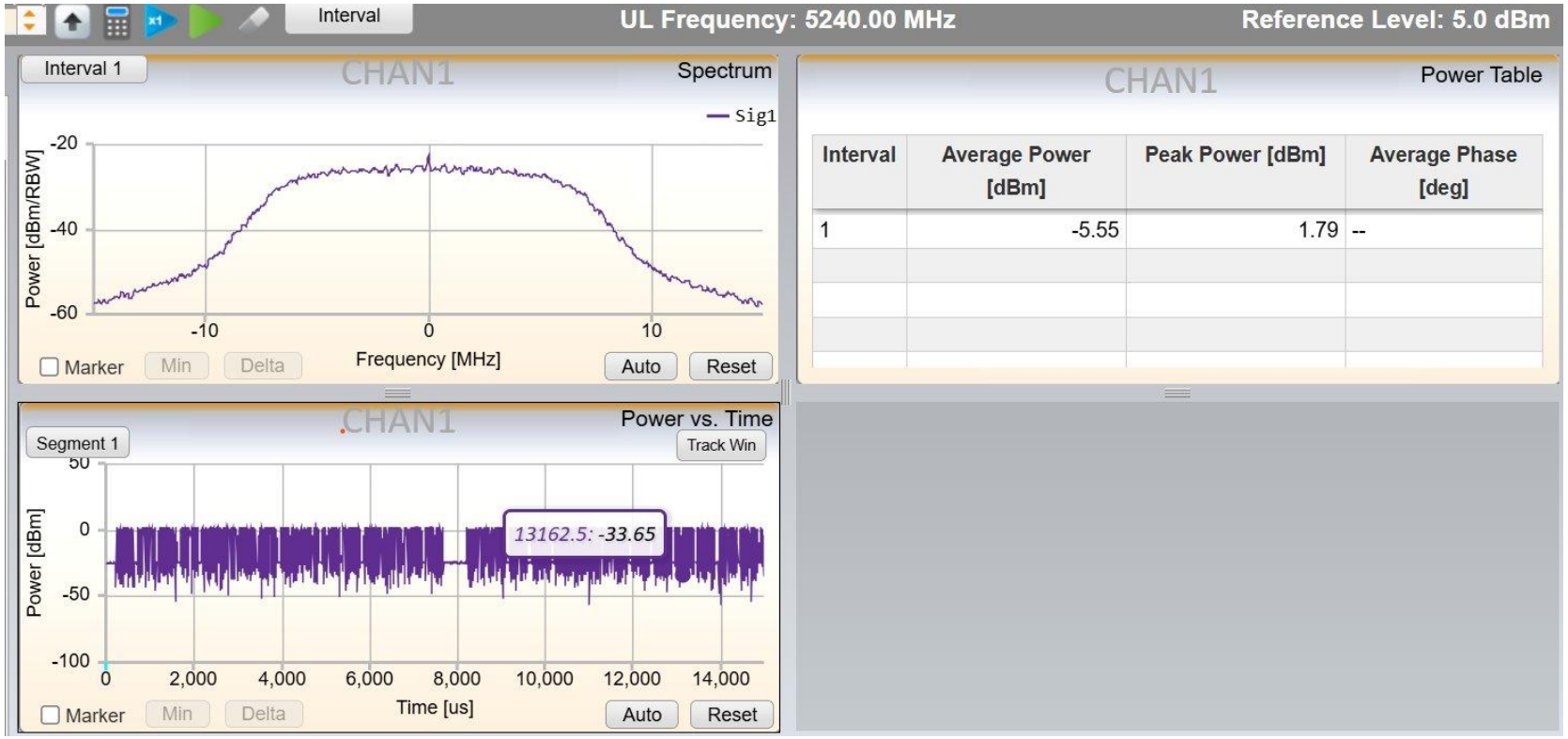
Output VSWR



TYPICAL INSERTION LOSS	2.4 GHz	5 GHz	7 GHz
Specified values	6 dB	7 dB	8.5 dB

Specified accuracy ±0.5 dB. Switching time 2 µs. Control/power options include PoE, Ethernet, USB and REST.

20 MHz AWGN stimulus



5240

MHz center frequency

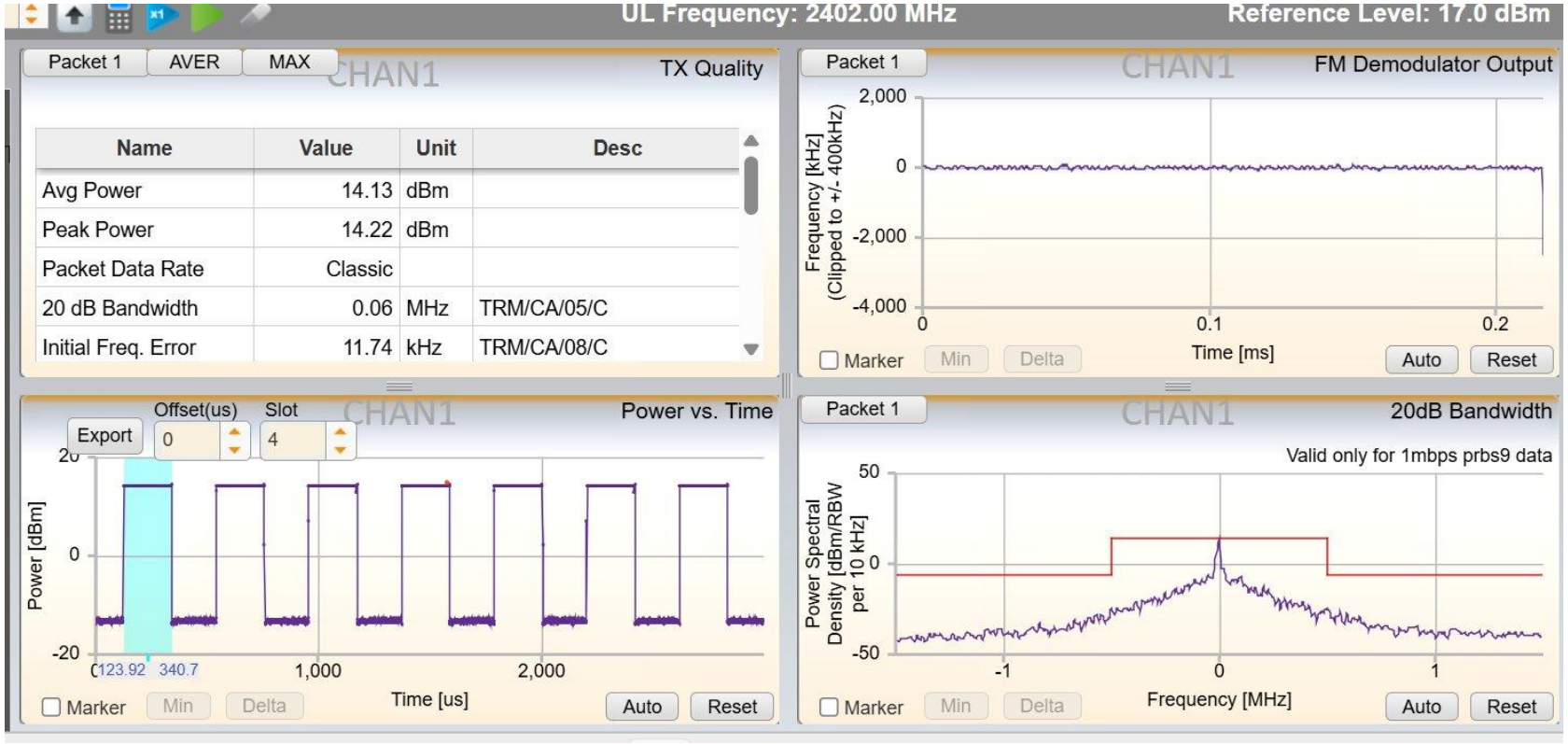
−5.55

dBm average power shown

Receiver desense and blocking

The spectrum and power-versus-time views characterize this stimulus. Delivered DUT power includes the intervening RF path.

Bluetooth Classic BR measurement



2402

MHz center frequency

14.13

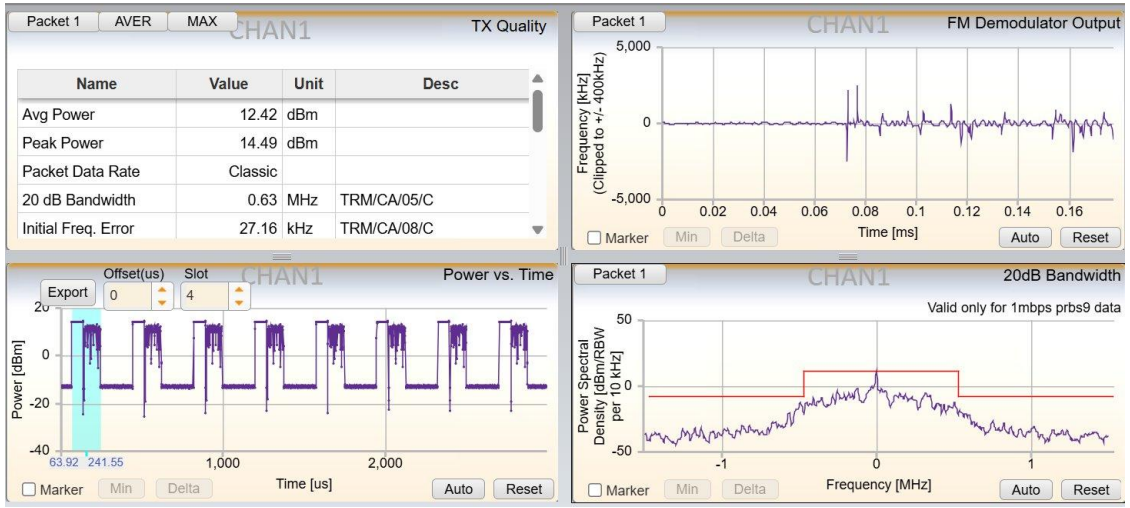
dBm average power shown

1 Mbps GFSK stimulus

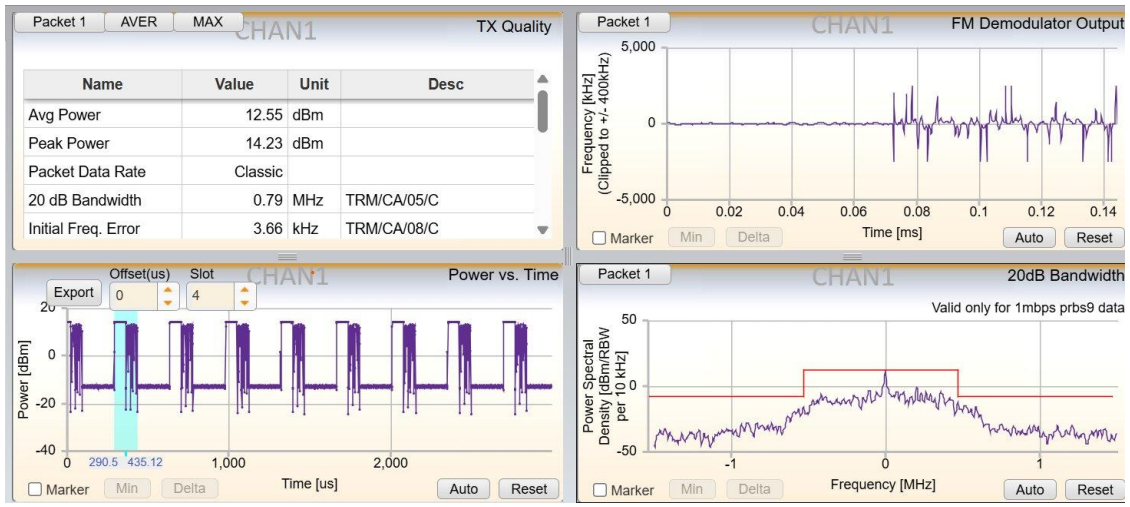
The capture shows burst timing and spectral behavior. Formal modulation and mask assessment requires the appropriate test mode.

Bluetooth EDR: 2 Mbps and 3 Mbps examples

2 Mbps, π/4-DQPSK



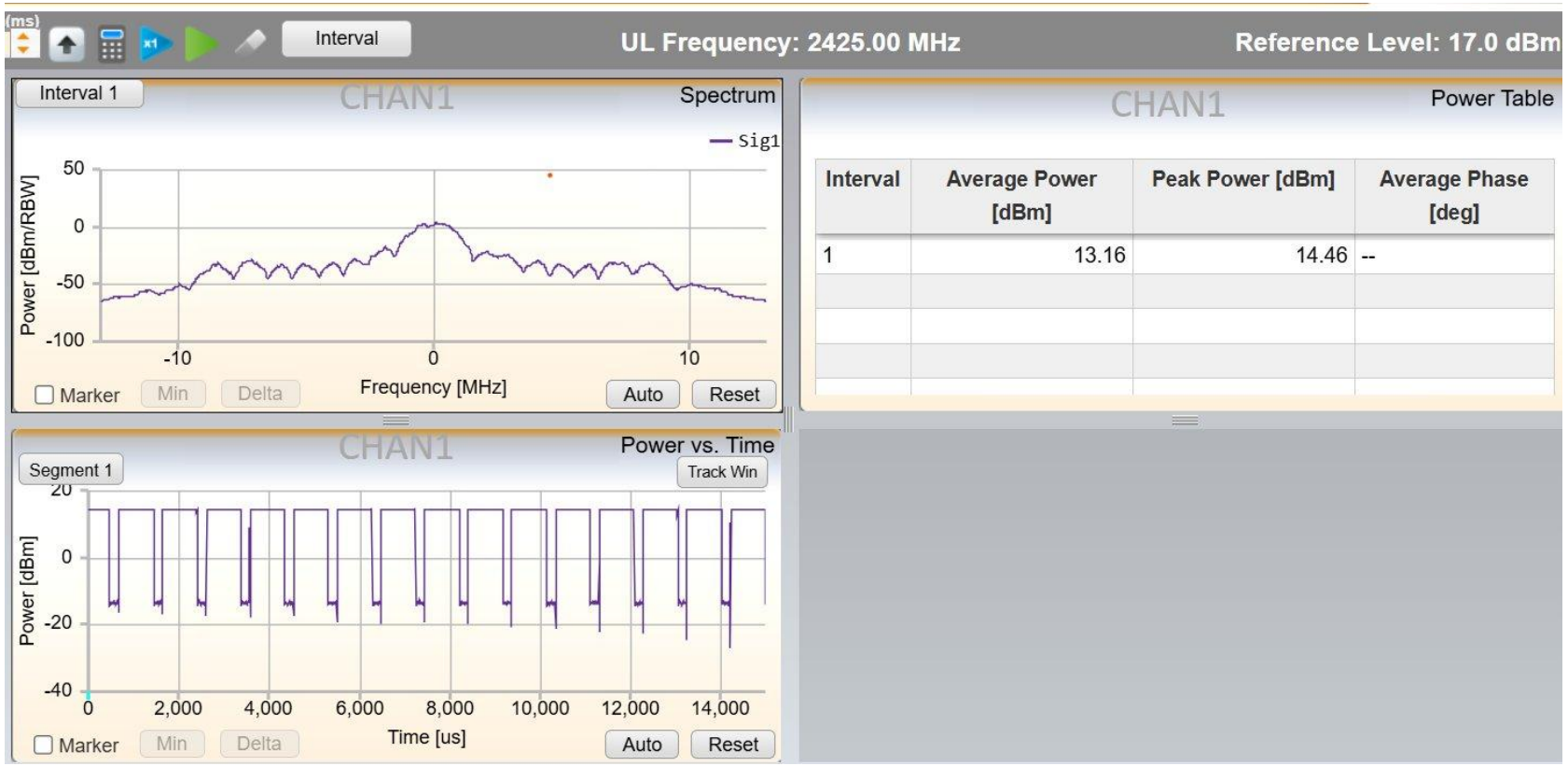
3 Mbps, 8DPSK



MEASUREMENT SHOWN	2 Mbps EXAMPLE	3 Mbps EXAMPLE
Average power	12.42 dBm	12.55 dBm

Representative EDR waveform views. Dedicated EDR measurements are required for modulation-quality or conformance conclusions.

Zigbee: IEEE 802.15.4 coexistence stimulus



2425

MHz, channel 15

13.16

dBm average power shown

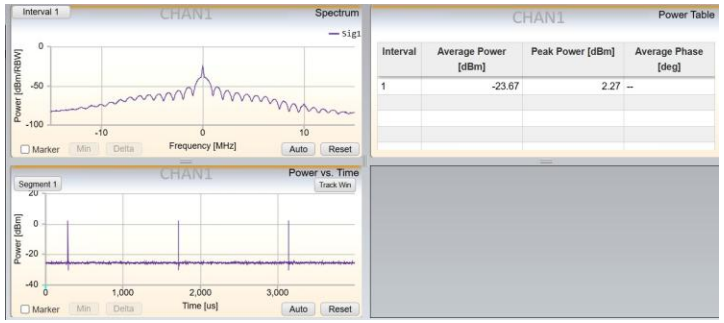
O-QPSK with half-sine chip shaping

Original spectrum and burst views support stimulus characterization for 2.4 GHz coexistence tests.

FCC radar waveform examples

Representative instrument views at 5260 MHz for six radar profiles.

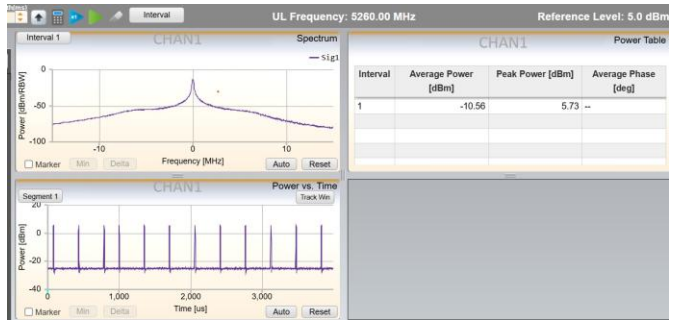
Type 1



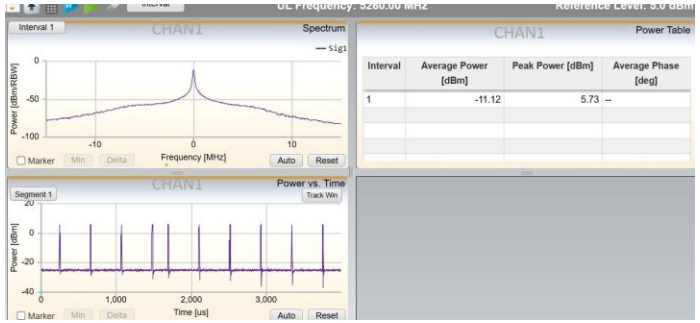
Type 2



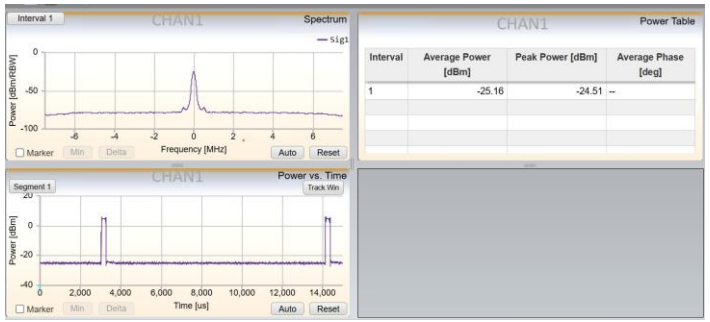
Type 3



Type 4



Type 5



Type 6

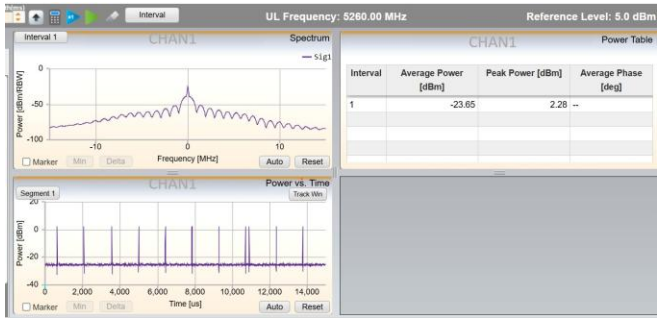


Short-pulse types 1–4, long-pulse type 5 and hopping type 6. Individual views do not prove the full timing/hopping sequence or certification.

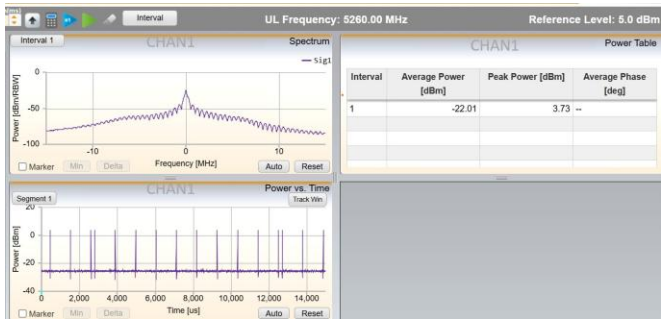
ETSI radar waveform examples

Representative instrument views at 5260 MHz for six radar profiles.

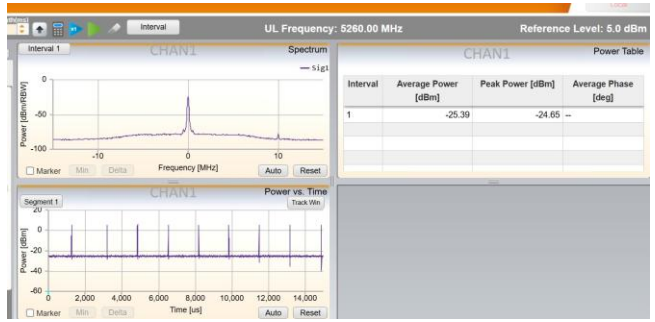
Type 1



Type 2



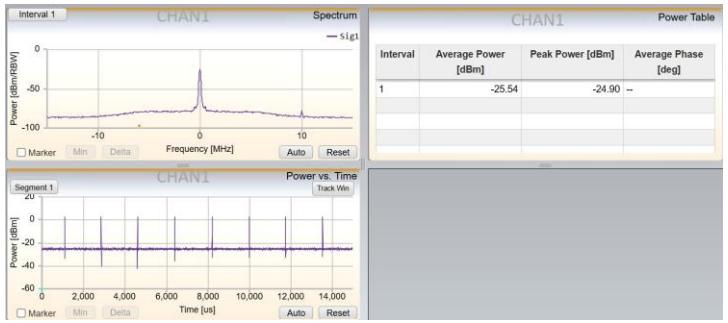
Type 3



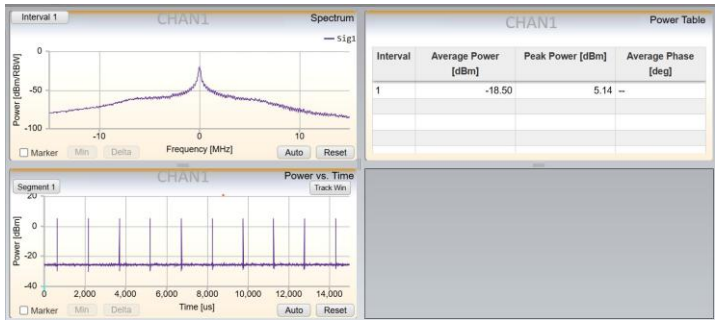
Type 4



Type 5



Type 6



Types 1–3: constant PRF. Type 4: chirp. Types 5–6: staggered PRF. Select the applicable EN 301 893 edition and test parameters.

A test configuration for your DUT

Client scale, RF paths and measurements matched to your validation priorities

support@wavemetrik.com

wavemetrik.com

Wavemetrik Technologies
Bengaluru, India